

A Universal Quantum Resource Theory of the Non-Hermitian Skin Effect: Position-Resolved Biorthogonality, Reciprocal Free Objects, and Exact Conversion Cost

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Abstract

We construct a quantum resource theory of the non-Hermitian skin effect (NHSE) whose resource is *geometric*, not merely a nonnormality scalar. The object is the biorthogonal atlas of a generator on a finite register of cells; each eigenmode contributes a pair of pure states (right, left). Two monotones are built from one primitive, the order-1/2 fidelity divergence: the nonnormality budget Φ_Q , monotone under all CPTP maps, and the *skin budget* $\Phi_X \leq \Phi_Q$, the same divergence between the cell-position densities of right and left modes, monotone under position-covariant maps. We prove: Φ_X vanishes exactly on reciprocal objects (on-site transposition symmetry) and on every translation-invariant ring, even when $\Phi_Q > 0$ there, so the periodic reference is exactly free; nonlocal unitaries are not free, which is why Chien’s Toeplitz-to-symmetric similarity does not trivialize the theory; free operations are position-covariant, boundary-blind, reference-sufficient, and reference-sufficiency is shown to be necessary by an explicit counterexample; conversion under free maps is governed by Blackwell’s theorem; a real-space theorem gives an extensive resource without translation invariance; the Hatano–Nelson chain is solved exactly, $\Phi = |g|(L-1) - \log(L+1) + \log(\frac{1}{2} \coth |g|) + \frac{2}{L} \log[(L+1)(1-e^{-2|g|})] + O(e^{-2|g|L})$; and the minimal conditioning cost of dressing one HN skin sector into another equals the resource difference up to $O(1)$. A claims ledger separates what is proved, imported, conditional, conjectural, and open.

1 Introduction, scope, and what “universal” means

The NHSE is defined by a boundary contrast: one bulk rule gives different spectra and eigenmodes under open (OBC) and periodic (PBC) boundaries [1, 2, 3]. Nakai et al. established that OBC skin accumulation produces a macroscopic enhancement of nonnormality and introduced eigenvector condition-number diagnostics [4]; the Petermann factor $K_\lambda = |\langle \ell_\lambda | r_\lambda \rangle|^{-2}$ and Wilkinson’s eigenvalue condition number are classical [12, 13]. We therefore do *not* claim that exponential nonnormality is new. The question is:

Which quantum object carries the skin resource, which operations are free, which quantity is monotone, and what does it cost to change the resource?

Two obstructions and their resolution. (i) Every finite complex Toeplitz matrix is unitarily similar to a complex symmetric one [6]. Hence any resource invariant under *all* unitaries, in particular any function of $\|P_\lambda\|$ alone, cannot distinguish the OBC Hatano–Nelson (HN) chain

from a reciprocal system. The resolution is that the skin resource must be *position-resolved*: it lives in the cell densities of right and left modes, and the free unitaries are only those that respect the cell register. (ii) The difference of two monotones is not a monotone. The resolution is to prove monotonicity in the class where it is true, to show that the extra condition is automatic for translation-invariant references, and to prove by counterexample that it cannot be dropped in general.

Meaning of “universal”. We claim universality in four precisely stated senses and no others.

- U1 Domain.** The theory is defined for every finite-dimensional generator with simple spectrum on a register of cells $X \times [q]$, for any lattice, dimension, disorder, Fock-space register, or vectorized Liouvillian; no translation invariance or GBZ is used.
- U2 Structure.** One object class, one free-operation class, a family of monotones, and (via Blackwell) a complete conversion criterion at the atlas level.
- U3 Free set.** Reciprocal objects and translation-invariant rings are exactly free (Theorem 1).
- U4 Calibration.** An exact solution (HN) and an extensivity theorem for macroscopically separated modes.

Not claimed: a universal scaling law, a proof that $r_{\text{skin}} > 0$ characterizes every NHSE phase (Conjecture 1), or a dynamical-level (instrument) monotone (Open problem 1). Section 11 lists every item with its status.

2 The quantum object

Definition 1 (Geometric generator and boundary pair). *Fix a finite register of cells X , $|X| = L$, and q orbitals per cell, so $\mathcal{H} = \mathbb{C}^X \otimes \mathbb{C}^q$ with cell projectors Π_x . A geometric generator is any $H \in \mathbb{C}^{N \times N}$, $N = qL$, with simple spectrum. A boundary pair is (H_O, H_P) on the same register; for a bulk rule $H(\beta) = \sum_{a=-R_-}^{R_+} H_a \beta^a$ they are the block-Toeplitz and block-circulant matrices. Disordered rules are allowed: (H_O, H_P) are two prescribed realizations of one device family.*

Definition 2 (Biorthogonal atlas and dichotomy family). *For each eigenvalue: $H r_\lambda = \lambda r_\lambda$, $H^\dagger \ell_\lambda = \bar{\lambda} \ell_\lambda$, $\|r_\lambda\| = \|\ell_\lambda\| = 1$. The atlas is $\mathcal{A}(H) = \{\lambda; \rho_\lambda^R, \rho_\lambda^L\}$ with $\rho_\lambda^R = |r_\lambda\rangle\langle r_\lambda|$, $\rho_\lambda^L = |\ell_\lambda\rangle\langle \ell_\lambda|$. The cell-measurement channel $\Delta_X(\rho) = \sum_x \text{Tr}(\Pi_x \rho) |x\rangle\langle x|$ produces the cell densities*

$$p_\lambda(x) = \|\Pi_x r_\lambda\|^2, \quad q_\lambda(x) = \|\Pi_x \ell_\lambda\|^2.$$

The dichotomy family is $\mathcal{D}(H) = \{(p_\lambda, q_\lambda)\}_\lambda$.

Remark 1 (Physical preparation of the atlas states). $K_H(t) = e^{-i(H-i\gamma)t}$ is a contractive no-jump Kraus operator for suitable γ (a valid quantum instrument together with its failure branch). Evolving a generic input under $K_H(t)$ and postselecting on no jump prepares the right state of the eigenvalue with largest imaginary part; evolving under $H^\dagger$ prepares the left state; spectral filtering selects other modes. Thus both members of each pair are laboratory states, and the “discrimination of right from left” below compares the outputs of the forward and adjoint dynamics. The theory is defined on the atlas; we make no claim that a map acting on the atlas is an operation on the dynamics (Open problem 1).

3 Two budgets from one primitive

The primitive is the order-1/2 root-fidelity divergence $D_{1/2}(\rho\|\sigma) = -2 \log F(\rho, \sigma)$, $F = \|\sqrt{\rho}\sqrt{\sigma}\|_1$; this is the sandwiched Rényi divergence of order 1/2 for all states, and coincides with the Petz one for pure states.

Definition 3 (Budgets). *For simple spectrum and uniform mode weights,*

$$b_\lambda = -\log |\langle \ell_\lambda | r_\lambda \rangle| = \frac{1}{2} D_{1/2}(\rho_\lambda^R \| \rho_\lambda^L), \quad \Phi_Q(H) = \frac{1}{N} \sum_\lambda b_\lambda, \quad (1)$$

$$\beta_\lambda = -\log \text{BC}(p_\lambda, q_\lambda) = \frac{1}{2} D_{1/2}(\Delta_X \rho_\lambda^R \| \Delta_X \rho_\lambda^L), \quad \Phi_X(H) = \frac{1}{N} \sum_\lambda \beta_\lambda, \quad (2)$$

where $\text{BC}(p, q) = \sum_x \sqrt{p(x)q(x)}$ is the Bhattacharyya coefficient.

Proposition 1 (Identities and chain). (a) $\|P_\lambda\| = e^{b_\lambda} = K_\lambda^{1/2}$ with $P_\lambda = |r_\lambda\rangle\langle\ell_\lambda|/\langle\ell_\lambda|r_\lambda\rangle$. (b) $|\langle\ell_\lambda|r_\lambda\rangle| \leq F(\Delta^{\text{pin}}\rho^R, \Delta^{\text{pin}}\rho^L) \leq \text{BC}(p_\lambda, q_\lambda) \leq 1$, where Δ^{pin} is the cell pinching. Hence $b_\lambda \geq \beta_\lambda \geq 0$ and $\Phi_Q \geq \Phi_X \geq 0$. (c) $\beta_\lambda = 0$ iff $p_\lambda = q_\lambda$.

Proof. (a) The rank-one projector has norm $\|r\| \|\ell\| / |\langle\ell|r\rangle|$. (b) The first inequality is data processing for F under pinching; for the second, $F = \sum_x \|\Pi_x r\| \|\Pi_x \ell\| |\langle\hat{\ell}_x|\hat{r}_x\rangle|$ with unit vectors $\hat{r}_x, \hat{\ell}_x$ in cell x , and $|\langle\hat{\ell}_x|\hat{r}_x\rangle| \leq 1$. (c) $\sum_x (\sqrt{p} - \sqrt{q})^2 = 2(1 - \text{BC})$. $\square$

Theorem 1 (Properties of the budgets; status: proved). *Let H have simple spectrum.*

- (a) Faithfulness of Φ_Q : $\Phi_Q = 0 \iff H$ normal. Faithfulness of Φ_X : $\Phi_X = 0 \iff p_\lambda = q_\lambda$ for all λ .
- (b) Reciprocity is free. If there is a unitary V commuting with all Π_x such that $VH^T V^\dagger = H$ (on-site transposition symmetry; e.g. $H = H^T$), then $\Phi_X(H) = 0$, although $\Phi_Q(H)$ may be positive.
- (c) Translation-invariant rings are free. If H_P commutes with the cyclic cell translation $\mathcal{T}$, then $\Phi_X(H_P) = 0$ exactly, for every q , even when $\Phi_Q(H_P) > 0$.
- (d) Unitary control cannot exceed the nonnormality budget. For every unitary U , $0 \leq \Phi_X(UHU^\dagger) \leq \Phi_Q(H)$; Φ_Q is unitarily invariant.
- (e) Additivity. For $H_1 \otimes \mathbf{1} + \mathbf{1} \otimes H_2$ with simple spectrum and product register, Φ_Q and Φ_X are additive.
- (f) Separation. If $\Phi_X(H) \geq s$ then some mode has $\text{TV}(p_\lambda, q_\lambda) \geq 1 - e^{-s}$.

Proof. (a) If $\Phi_Q = 0$ then $\ell_\lambda \propto r_\lambda$, so r_λ is an eigenvector of both H and $H^\dagger$; then $(\lambda - \mu)\langle r_\mu | r_\lambda \rangle = 0$ and the eigenbasis is orthonormal, so H is normal; the converse is immediate. The Φ_X statement is Prop. 1(c) applied mode by mode. (b) From $H^T = V^\dagger H V$ we get $H^\dagger = \overline{H^T} = V^T \bar{H} \bar{V}$. Then $H^\dagger \ell = \bar{\lambda} \ell$ reads $\bar{H}(\bar{V} \ell) = \bar{\lambda} \bar{V} \ell$, i.e. $H(V \bar{\ell}) = \lambda V \bar{\ell}$. By simplicity $r_\lambda \propto V \bar{\ell}_\lambda$. Since V commutes with every Π_x , $\|\Pi_x r_\lambda\| = \|\Pi_x \ell_\lambda\|$. (c) With simple spectrum each r_λ is a $\mathcal{T}$ -eigenvector; $\mathcal{T}$ permutes cells unitarily, so $\|\Pi_{x+1} r\| = \|\Pi_{x+1} \mathcal{T} r\| = \|\mathcal{T} \Pi_x r\| = \|\Pi_x r\|$: p_λ is uniform. The same holds for ℓ_λ since $\mathcal{T}$ commutes with $H^\dagger$. So $p = q = 1/L$. (d) $UHU^\dagger$ has eigenvectors $Ur, U\ell$, so Φ_Q is invariant; then apply Prop. 1(b). (e) Eigenvectors and biorthogonal overlaps are tensor products, BC multiplies on product densities, and the uniform average over product labels splits. (f) $\max_\lambda \beta_\lambda \geq \Phi_X \geq s$ gives $\text{BC} \leq e^{-s}$, and $\text{TV} \geq 1 - \text{BC}$. $\square$

Remark 2 (Why nonlocal unitaries are not free; resolution of the Toeplitz obstruction). By [6] the OBC HN matrix is unitarily similar to a complex symmetric matrix S_{sym} via a nonlocal unitary U . By (d), $\Phi_Q(S_{\text{sym}}) = \Phi_Q(HN)$, but in the basis of S_{sym} Theorem 1(b) gives $\Phi_X = 0$. Skin content is basis-dependent by design: nonlocal unitaries can convert skin into unlocalized nonnormality. Numerically (Sec. 10, C7) a random unitary leaves Φ_Q unchanged and reduces Φ_X . The free unitaries are the position-covariant ones (Prop. 3).

Proposition 2 (Operational meaning; status: proved). (a) Sensitivity. To first order, $\delta\lambda = \langle \ell | \delta H | r \rangle / \langle \ell | r \rangle$, so $\sup_{\|\delta H\| \leq \varepsilon} |\delta\lambda| = \varepsilon e^{b_\lambda}$, attained by $\delta H = \varepsilon |\ell\rangle\langle r|$. For $\delta H = \Pi_A \delta H \Pi_B$ supported on cell sets A, B : $|\delta\lambda| \leq \varepsilon e^{b_\lambda} \sqrt{q_\lambda(A)p_\lambda(B)}$. (b) Discrimination. The quantum Chernoff exponent for discriminating ρ_λ^R from ρ_λ^L is $2b_\lambda = D_{1/2}(\rho_\lambda^R \| \rho_\lambda^L)$. The classical Chernoff exponent of the position statistics (p_λ, q_λ) is $\geq \beta_\lambda$, with equality for every HN mode; position measurement therefore retains exactly half of the optimal exponent for HN.

Proof. (a) is first-order perturbation theory and Cauchy–Schwarz. (b) For pure states $\text{Tr } \rho^s \sigma^{1-s} = |\langle \ell | r \rangle|^2$ for all s , so the quantum Chernoff exponent [14] is $-2 \log |\langle \ell | r \rangle|$. The classical exponent is $-\log \min_s \sum_x p^s q^{1-s} \geq -\log \sum_x \sqrt{p q} = \beta$, since $s = \frac{1}{2}$ is a competitor. For HN, Theorem 6 gives $q_j(L+1-n) = p_j(n)$, so $s \mapsto \sum p^s q^{1-s}$ is convex and symmetric about $\frac{1}{2}$, hence minimal at $s = \frac{1}{2}$. $\square$

The physical reading of Φ_X : the skin budget is the log-sensitivity of an eigenvalue to a perturbation that couples the two regions where right and left modes live; the PBC-closing link is exactly such a perturbation.

4 The NHSE resource theory

4.1 Objects and resource

Definition 4 (Boundary-paired object; resource). $X_L = (\mathcal{A}(H_O), \mathcal{A}(H_P))$. Signed excess $\Delta_B^{(L)} = \Phi_X(H_O) - \Phi_X(H_P)$; finite-size resource $R_{\text{skin}}^{(L)} = [\Delta_B^{(L)}]_+$; density $r_{\text{skin}} = [\limsup_L \Delta_B^{(L)} / L]_+$; the full sequence is the resource profile. A pair is free at size L if $R_{\text{skin}}^{(L)} = 0$, and a family is free if $R_{\text{skin}}^{(L)} = O(1)$.

By Theorem 1(c), for every translation-invariant bulk rule with simple PBC spectrum, $\Phi_X(H_P) = 0$ and $R_{\text{skin}}^{(L)} = \Phi_X(H_O)$: the periodic calibration is exactly free. (This corrects V10: PBC nonnormality, which can be extensive for multiband rules, sits in $\Phi_Q(H_P)$ and does not enter Φ_X .) The calibration term matters only for disordered or interacting references, where the ring may itself carry cell-density asymmetry.

4.2 Free operations

Definition 5 (Position-covariant, boundary-blind, reference-sufficient). A CPTP map Λ (possibly with ancillas in product states) is position-covariant if there is a classical channel T with $\Delta_X \circ \Lambda = T \circ \Delta_X$. A free operation Θ applies the same covariant Λ , with the same mode labels (bijective relabelings allowed, coarse-graining of modes not), to both members of X_L . It is reference-sufficient if the induced T is sufficient for the reference densities: there is a classical channel $\tilde{T}$ with $\tilde{T} T p_\lambda^P = p_\lambda^P$ and $\tilde{T} T q_\lambda^P = q_\lambda^P$ for all λ . No condition involves the value of R_{skin} .

Then $\Delta_X(\Lambda\rho_\lambda^R) = Tp_\lambda$ and $\Delta_X(\Lambda\rho_\lambda^L) = Tq_\lambda$: covariant maps act on the dichotomy family by a common stochastic matrix.

Proposition 3 (Richness and the free unitaries; status: proved). *The covariant class contains: measure-and-prepare maps from the cell measurement, cell-local CPTP maps, random cell permutations, dephasing, product ancillas. A unitary is covariant iff it is a cell permutation composed with an on-site unitary, $U = \sum_x |\pi(x)\rangle\langle x| \otimes V_x$. Reference-sufficiency holds whenever T is injective on the union of supports of the reference densities, in particular for all such unitaries.*

Proof. The listed maps satisfy $\Delta_X\Lambda = T\Delta_X$ by inspection (for measure-and-prepare, $T(x'|x) = \text{Tr } \Pi_{x'}\sigma_x$). For a unitary, covariance requires $\|\Pi_y U\psi\|^2$ to depend only on $\{\|\Pi_x\psi\|^2\}$. Taking ψ inside one cell forces $(\Pi_y U)^\dagger(\Pi_y U) \propto \mathbf{1}$ on each $V_x = \Pi_x\mathcal{H}$; taking ψ a superposition of two cells forces $\Pi_y UV_x \perp \Pi_y UV_{x'}$ for $x \neq x'$. A dimension count ($\sum_x \dim \Pi_y UV_x \leq q$ for each y , while $\sum_y \dim \Pi_y UV_x \geq q$) then forces each UV_x into a single cell. Injectivity of T on the supports gives a left inverse. $\square$

Local coherent unitaries such as a beam splitter between neighboring cells are *not* free: they can convert nonnormality into skin (e.g. $r = (1, 1)/\sqrt{2}$, $\ell = (1, -1)/\sqrt{2}$ have $\text{BC} = 1$; after the beam splitter $\text{BC} = 0$). This is the price of a geometric monotone, and Theorem 1(d) bounds what any unitary can create by Φ_Q .

Theorem 2 (Monotonicity of the NHSE resource; status: proved). *Let Θ be a boundary-blind covariant free operation. Then Φ_Q and Φ_X of each member satisfy $\Phi_Q(\Lambda\mathcal{A}) \leq \Phi_Q(\mathcal{A})$ for every CPTP Λ , and $\Phi_X(\Theta\mathcal{A}) \leq \Phi_X(\mathcal{A})$ for covariant Λ . For the boundary excess:*

- (a) *if the reference is reciprocal ($p_\lambda^P = q_\lambda^P$ for all λ ; this holds for every translation-invariant ring), then $R_{\text{skin}}^{(L)}(\Theta X_L) \leq R_{\text{skin}}^{(L)}(X_L)$ for every boundary-blind covariant Θ ;*
- (b) *in general, the same holds for every reference-sufficient Θ ;*
- (c) *consequently $r_{\text{skin}}(\Theta X) \leq r_{\text{skin}}(X)$ and the free sector is invariant.*

Proof. Classical data processing for BC: $\text{BC}(Tp, Tq) \geq \text{BC}(p, q)$ for any stochastic T (joint concavity of $\sqrt{pq}$, or the Cauchy–Schwarz form $\sum_{x'} \sqrt{\sum_x T_{x'x} p_x \sum_x T_{x'x} q_x} \geq \sum_x \sqrt{p_x q_x}$). Averaging over modes gives Φ_X monotonicity; Φ_Q is data processing of F . (a) $p^P = q^P \Rightarrow Tp^P = Tq^P$, so $\Phi_X(\Theta P) = 0 = \Phi_X(P)$ and $R'_{\text{skin}} = \Phi_X(\Theta O) \leq \Phi_X(O) = R_{\text{skin}}$. (b) Sufficiency gives $\Phi_X(P) = \Phi_X(\tilde{T}TP) \leq \Phi_X(TP) \leq \Phi_X(P)$, hence equality, and $R'_{\text{skin}} = [\Phi_X(\Theta O) - \Phi_X(P)]_+ \leq R_{\text{skin}}$. (c) divide by L , take $\limsup$; if $R_{\text{skin}}^{(L)} = O(1)$ the image is $O(1)$. $\square$

Proposition 4 (Reference-sufficiency is necessary in general; status: proved). *Let $X = \{1, 2, 3, 4\}$, one mode, O : $p = (\frac{3}{4}, \frac{1}{4}, 0, 0)$, $q = (\frac{1}{4}, \frac{3}{4}, 0, 0)$; P : $p = (0, 0, \frac{9}{10}, \frac{1}{10})$, $q = (0, 0, \frac{1}{10}, \frac{9}{10})$. Then $\Phi_X(O) = -\log \frac{\sqrt{3}}{2} \approx 0.144$, $\Phi_X(P) = -\log 0.6 \approx 0.511$, so $R_{\text{skin}} = 0$. The boundary-blind covariant T merging cell 4 into cell 3 gives $\Phi_X(TP) = 0$, $\Phi_X(TO) = 0.144$: R_{skin} increases from 0 to 0.144.*

By Theorem 1(c) such a reference cannot be a translation-invariant ring; the sufficiency condition is needed only for genuinely nonreciprocal references (disordered or interacting rings), and there it cannot be removed.

Theorem 3 (Complete conversion criterion at the atlas level; status: imported). *Let $\mathcal{D} = \{p_m\}_{m=1}^{2N}$ and $\mathcal{D}' = \{p'_m\}$ be the lists of all right and left cell densities of two atlases. A common stochastic map with $Tp_m = p'_m$ for all m exists iff for every finite decision problem the Bayes risk based on $\mathcal{D}$ is at most that based on $\mathcal{D}'$ (Blackwell's theorem [11]); for a single mode (2 hypotheses) this is Lorenz-curve dominance of (p, q) over (p', q') , equivalently monotonicity of all f -divergences.*

Thus the family {all f -divergence averages}, of which Φ_X is one member, is a complete set of monotones for single-mode atlas conversion by free maps. We make no claim about which HN conversions are free in this sense (Open problem 2).

Theorem 4 (Resource-theory axioms; status: proved). *(i) Objects: boundary-paired atlases. (ii) Free objects: $R_{\text{skin}}^{(L)} = 0$ (family: $O(1)$). (iii) Free operations: boundary-blind, covariant, reference-sufficient; closed under composition, tensoring with product ancillas, and mode relabeling (a composition in which each map is sufficient for the reference it acts on is sufficient, with recovery $\tilde{T}_1\tilde{T}_2$). (iv) Free objects are mapped to free objects (Theorem 2). (v) $R_{\text{skin}}^{(L)}$, r_{skin} are monotones, additive across independent boundary pairs with reciprocal references.*

5 Real-space extensivity theorem

Assumption 1 (Macroscopic boundary separation). *There exist $\kappa, A > 0$, a sequence $f_L \in (0, 1]$ and mode sets S_L with $|S_L| = f_L N$ such that for all $\lambda \in S_L$, cells $1 \leq n \leq L$, orbitals a : $|r_\lambda(n, a)| \leq Ae^{-\kappa(L-n)}$ and $|\ell_\lambda(n, a)| \leq Ae^{-\kappa(n-1)}$.*

Lemma 1. *For $\lambda \in S_L$: $\text{BC}(p_\lambda, q_\lambda) \leq qLA^2e^{-\kappa(L-1)}$, hence $\beta_\lambda \geq \kappa(L-1) - \log(qLA^2)$.*

Proof. $\|\Pi_n r\| \leq \sqrt{q}Ae^{-\kappa(L-n)}$, $\|\Pi_n \ell\| \leq \sqrt{q}Ae^{-\kappa(n-1)}$; the product is $qA^2e^{-\kappa(L-1)}$ for every n ; sum over L cells. $\square$

Theorem 5 (Extensive skin resource; status: proved). *Under Assumption 1,*

$$R_{\text{skin}}^{(L)} \geq f_L [\kappa(L-1) - \log(qLA^2)] - \Phi_X(H_P),$$

so $r_{\text{skin}} \geq \kappa \liminf f_L - \limsup \Phi_X(H_P)/L$. If H_P is translation-invariant, $\Phi_X(H_P) = 0$, and if $f_L \geq f - C/L$ then $R_{\text{skin}}^{(L)} \geq f\kappa L - O(\log L)$.

Proof. All $\beta_\lambda \geq 0$; modes in S_L contribute at least the Lemma bound; divide by N . This corrects V10, whose $o(1)$ error term should have been $o(L)$ when $|S_L| = fN + o(L)$. $\square$

Proposition 5 (Exact no-skin direction for Toeplitz rules; status: proved). *If the bulk rule satisfies $H_{-a} = VH_a^T V^\dagger$ for a fixed unitary V on $\mathbb{C}^q$ (for $q = 1$: $a_{-j} = a_j$, i.e. the symbol satisfies $f(\beta) = f(1/\beta)$), then H_O has on-site transposition symmetry, $\Phi_X(H_O) = 0$ and $R_{\text{skin}}^{(L)} = 0$ for all L , even if $\Phi_Q(H_O) > 0$.*

Conjecture 1 (Faithfulness for banded Toeplitz rules). *For $q = 1$, finite-range symbols with simple OBC spectrum: $r_{\text{skin}} > 0$ iff the point-gap winding of the symbol is nonzero at some point of the limiting OBC spectrum (equivalently the GBZ is not the unit circle). The direction “reciprocal $\Rightarrow r_{\text{skin}} = 0$ ” is the Proposition above; the converse is expected to follow from Schmidt–Spitzer/Reichel–Trefethen asymptotics of banded Toeplitz spectra and eigenvector conditioning [16, 17] but is not proved here.*

6 Hatano–Nelson: exact calibration

Let $H_{g,L}^O = m\mathbf{1} + e^g T_+ + e^{-g} T_-$, $T_+ = \sum_{n=1}^{L-1} |n+1\rangle\langle n|$, $T_- = T_+^T$, $m, g \in \mathbb{R}$; the case $g < 0$ follows by reflection, so take $g > 0$, $u = e^{-2g}$, $k_j = \pi j / (L+1)$, $s_j(n) = \sin k_j n$. Then $r_j(n) = e^{g(n-1)} s_j(n) / R_j$, $\ell_j(n) = e^{-g(n-1)} s_j(n) / L_j$ with $R_j^2 = \sum_n e^{2g(n-1)} s_j^2$, $L_j^2 = \sum_n e^{-2g(n-1)} s_j^2$.

Theorem 6 (HN calibration; status: proved). (i) $R_j = e^{g(L-1)} L_j$ and $q_j(L+1-n) = p_j(n)$ exactly; $\langle \ell_j | r_j \rangle = \text{BC}(p_j, q_j) = \frac{L+1}{2R_j L_j}$; hence $\Phi_X = \Phi_Q$ and $b_j = g(L-1) + \log \frac{2L_j^2}{L+1}$.

(ii) With $u = e^{-2g}$,

$$\Phi_Q(H_{g,L}^O) = g(L-1) - \log(L+1) + \log\left(\frac{1}{2} \coth g\right) + \frac{2}{L} \log[(L+1)(1-u)] - \frac{2}{L} \log(1-u^{L+1}) + O_g(e^{-2gL}).$$

In particular $\Phi_Q = g(L-1) - \log(L+1) + \log(\frac{1}{2} \coth g) + O(\frac{\log L}{L})$ and, within the elementary bounds, $-\log 2 \leq \Phi_Q - g(L-1) + \log(L+1) \leq \log \frac{2}{1-u}$.

(iii) For $g \neq 0$ the ring has simple spectrum ($k \neq -k$) and $\Phi_Q(H_{g,L}^P) = \Phi_X(H_{g,L}^P) = 0$.

(iv) Hence $R_{\text{skin}}^{(L)} = |g|(L-1) - \log(L+1) + \log(\frac{1}{2} \coth |g|) + o(1)$ and $r_{\text{skin}} = |g|$.

Proof. (i) Since $\sin^2 k_j(L+1-n) = \sin^2 k_j n$, substituting $n \rightarrow L+1-n$ in L_j^2 gives $L_j^2 = e^{-2g(L-1)} R_j^2$ and $q_j(L+1-n) = p_j(n)$. Because $r_j(n)\ell_j(n) = s_j^2/(R_j L_j) \geq 0$, $\langle \ell_j | r_j \rangle = \sum_n \sqrt{p_j q_j} = \sum s_j^2/(R_j L_j) = (L+1)/(2R_j L_j)$ using $\sum_{n=1}^L s_j(n)^2 = (L+1)/2$. Then $b_j = \log(2R_j L_j/(L+1))$ and $R_j = e^{g(L-1)} L_j$ give the formula. (ii) Summing the geometric-trigonometric series to infinity, $\Lambda(k) = \sum_{n \geq 1} u^{n-1} \sin^2 kn = \frac{(1+u)\sin^2 k}{(1-u)(1-2u \cos 2k + u^2)}$. The truncation error is $\tau_j = u^L \sum_{m \geq 0} u^m \sin^2(k_j m)$ (using $k_j(L+1) = \pi j$), and with $\tilde{k} = \min(k_j, \pi - k_j)$, $\tau_j \leq u^L \tilde{k}^2 u(1+u)/(1-u)^3$ while $\Lambda(k_j) \geq 4\tilde{k}^2/(\pi^2(1-u^2))$; hence $\tau_j/\Lambda(k_j) \leq C_g u^L$ uniformly in j and $\log L_j^2 = \log \Lambda(k_j) + O(u^L)$. Now $\log \Lambda(k) = \log \frac{1+u}{1-u} + \log \sin^2 k - \log(1-2u \cos 2k + u^2)$, and exact identities give $\sum_{j=1}^L \log \sin^2 k_j = 2 \log \frac{L+1}{2L}$ and, from $\prod_{j=0}^L (1-2u \cos \frac{2\pi j}{L+1} + u^2) = (1-u^{L+1})^2$, $\sum_{j=1}^L \log(1-2u \cos 2k_j + u^2) = 2 \log(1-u^{L+1}) - 2 \log(1-u)$. Insert in (i) and use $\frac{1+u}{1-u} = \coth g$. The elementary bounds follow from $R_j L_j \geq e^{g(L-1)} \sin^2 k_j$ and $R_j^2 \leq e^{2g(L-1)}/(1-u)$, $L_j^2 \leq 1/(1-u)$ together with the exact average of $\log \sin^2 k_j$. (iii) The ring is circulant, hence unitarily diagonalized by the Fourier matrix; then $r = \ell$ (both plane waves) and all overlaps are 1. (iv) Definition of R_{skin} and (iii). $\square$

Remark 3 (Relation to Nakai et al.). *The leading rate coincides with the eigenvector condition-number growth of [4] ($e^{|g|} = r_N$ in their parametrization). The sharp constant $\log(\frac{1}{2} \coth |g|)$ and the $-\log(L+1)$ term are the new exact content; the numerics in Sec. 10 confirm the formula to 10^{-9} at $L = 40$.*

7 Regular finite-range sector: GBZ representation

The GBZ is a coordinate representation, not part of the definition.

Assumption 2 (Envelope hypothesis, for a translation-invariant rule). *Modes are labeled by $\lambda = (\alpha, \theta)$ with skin exponents $a_\lambda = |\log |\beta_\alpha(\theta)|| \leq a_{\max}$, and (orienting each mode so that right modes sit at large n) cell densities $p_\lambda(x) = e^{2a_\lambda x} u_\lambda(x) / Z_\lambda^p$, $q_\lambda(x) = e^{-2a_\lambda x} v_\lambda(x) / Z_\lambda^q$. There is $\varepsilon_L \rightarrow 0$ uniform in λ with: (E1) $u_\lambda, v_\lambda \leq e^{\varepsilon_L L}$; (E2) some $x \geq L - \varepsilon_L L$ has $u_\lambda(x) \geq e^{-\varepsilon_L L}$ and some $x' \leq \varepsilon_L L$ has $v_\lambda(x') \geq e^{-\varepsilon_L L}$; (E3) $\sum_x \sqrt{u_\lambda v_\lambda} \geq e^{-\varepsilon_L L}$. Mode equidistribution: $\frac{1}{N} \sum_\lambda a_\lambda \rightarrow \frac{1}{q} \sum_\alpha \int_0^{2\pi} \frac{d\theta}{2\pi} |\log |\beta_\alpha(\theta)||$.*

Theorem 7 (GBZ representation; status: conditional on Assumption 2). *Uniformly in λ , $|\beta_\lambda - a_\lambda L| \leq (2a_{\max} + 2)\varepsilon_L L + \log L$, hence*

$$r_{\text{skin}} = \frac{1}{q} \sum_{\alpha=1}^q \int_0^{2\pi} \frac{d\theta}{2\pi} |\log |\beta_\alpha(\theta)||.$$

Proof. BC = $\sum_x \sqrt{uv}/\sqrt{Z^p Z^q}$. Upper bound on BC: numerator $\leq Le^{\varepsilon L}$ by (E1); (E2) gives $Z^p \geq e^{2a(L-\varepsilon L)-\varepsilon L}$, $Z^q \geq e^{-2a\varepsilon L-\varepsilon L}$, so BC $\leq Le^{-aL+(2a+2)\varepsilon L}$. Lower bound: numerator $\geq e^{-\varepsilon L}$ by (E3); $Z^p \leq Le^{2aL+\varepsilon L}$, $Z^q \leq Le^{\varepsilon L}$, so BC $\geq e^{-aL-2\varepsilon L}/L$. Average over modes and use $\Phi_X(H_P) = 0$ (Theorem 1(c)). $\square$

The HN chain satisfies the hypothesis with $u = v = \sin^2$ and $a_\lambda = g$, reproducing $r_{\text{skin}} = |g|$. Generalized/relative skin effects that violate a single-GBZ description [10] violate only this representation, not the resource theory.

8 Cost of nonfree conversion

Free operations act on atlases. A different, physically important operation is *dressing* of the generator, $H' = SHS^{-1}$, which conjugates all dynamics and is implemented by postselected Kraus operators proportional to S and S^{-1} .

Proposition 6 (Dressing; status: proved). *(a) For any invertible S and $H' = SHS^{-1}$ with the modes matched, $r'_\lambda \propto Sr_\lambda$, $\ell'_\lambda \propto S^{-\dagger}\ell_\lambda$, $|b'_\lambda - b_\lambda| \leq \log \text{cond}(S)$; so $|\Phi_Q(H') - \Phi_Q(H)| \leq \log \text{cond}(S)$. (b) For cell-diagonal positive $S = \text{diag}(s_x)$: $p' = S^2 p/Z^p$, $q' = S^{-2} q/Z^q$ with $Z^p = \sum s_x^2 p_x$, $Z^q = \sum s_x^{-2} q_x$; then $\beta' = \beta + \frac{1}{2} \log(Z^p Z^q)$ exactly and $|\beta' - \beta| \leq \log \text{cond}(S)$; so $|\Phi_X(H') - \Phi_X(H)| \leq \log \text{cond}(S)$. (c) The Kraus operators $S/s_{\max}$ and $s_{\min} S^{-1}$ succeed with probability at least $\text{cond}(S)^{-2}$ on every state, with equality on a state in an extremal cell.*

Proof. (a) $P'_\lambda = SP_\lambda S^{-1}$, so $\|P'\| \leq \text{cond}(S)\|P\|$ and symmetrically. (b) $\text{BC}(p', q') = \sum \sqrt{p'q'}/\sqrt{Z^p Z^q}$, and $\sqrt{Z^p Z^q} \in [s_{\min}/s_{\max}, s_{\max}/s_{\min}]$. (c) Direct. $\square$

Theorem 8 (Exact dressing cost on the HN ray; status: proved). *Let $g_1, g_2 \neq 0$ and $\mathcal{S}(g_1, g_2) = \{S \text{ invertible} : SH_{g_1,L}^O S^{-1} = H_{g_2,L}^O\}$. Then*

- (i) (achievability) with W the cell reflection ($WH_g W^\dagger = H_{-g}$, unitary and free) and $S_\delta = \text{diag}(e^{-\delta(n-1)})$, an element of $\mathcal{S}$ with $\log \text{cond} = ||g_2| - |g_1|| (L-1)$ exists;
- (ii) (converse) every $S \in \mathcal{S}$ has $\log \text{cond}(S) \geq |\Phi_Q(g_2) - \Phi_Q(g_1)| = ||g_2| - |g_1|| (L-1) - c_L$ with $c_L \rightarrow \left| \log \frac{\coth |g_2|}{\coth |g_1|} \right|$;
- (iii) (operational form) for $g_1 > g_2 > 0$ the dressing S_δ , $\delta = g_1 - g_2$, converts every mode exactly ($r_j \mapsto r'_j$, $\ell_j \mapsto \ell'_j$) and every right/left branch succeeds with probability $e^{-2\delta(L-1)}$ times a factor between $c_g L^{-2}$ and $c'_g L^2$.

Therefore $\lim_L \frac{1}{L} \inf_{S \in \mathcal{S}} \log \text{cond}(S) = ||g_2| - |g_1||$: the minimal conditioning cost density equals the difference of resource densities, and $\inf \log \text{cond}$ lies within $O(1)$ of $||g_2| - |g_1|| (L-1)$. For a cost budget c (per site), the sector $g_1 \rightarrow g_2$ is reachable, in the density sense, iff $||g_2| - |g_1|| \leq c$.

Proof. (i) $S_\delta H_{g_1} S_\delta^{-1} = H_{g_1 - \delta}$ because $S_\delta T_+ S_\delta^{-1} = e^{-\delta} T_+$ and $S_\delta T_- S_\delta^{-1} = e^\delta T_-$; for opposite signs first apply W . $\text{cond}(S_\delta) = e^{|\delta|(L-1)}$ and $\text{cond}(SW) = \text{cond}(S)$. (ii) Similar matrices have equal spectra, so modes are matched by eigenvalue and Prop. 6(a) applies; then use Theorem 6(ii). (iii) $r_j \propto e^{g_1(n-1)} s_j$ maps to $e^{g_2(n-1)} s_j$; the success probabilities are $\sum_n e^{-2\delta(n-1)} p_j(n)$ and $e^{-2\delta(L-1)} \sum_n e^{2\delta(n-1)} q_j(n)$; bound numerator and denominator using $\sin^2 k_j \geq (2/(L+1))^2$ and geometric sums. $\square$

Remark 4 (Scope of the cost theorem). *The cost is defined for dressing similarities of the generator, with $\log \text{cond}$ as the conditioning (equivalently $\frac{1}{2} \log(1/\text{success probability})$ in the worst case). It is not a claim about all postselected channels and it is not a completeness statement for the HN ray: any scalar monotone is trivially complete on a one-parameter family. The content is the exact matching of cost and resource difference, and the fact that dressing conditioning is bounded below by the monotone.*

9 Sectors: what the definition covers and what is proved

The definition applies verbatim to every sector; only the theorems differ.

- *Regular finite skin depth:* Theorem 5 (proved), Theorem 7 (conditional).
- *Critical scale-free:* the resource profile $R_{\text{skin}}^{(L)}$ is defined; we set $\chi_{\text{skin}} = \limsup \log(1 + R_{\text{skin}}^{(L)}) / \log L$. Values of χ_{skin} are an open computation [8, 9].
- *Disorder:* Theorems 1, 2, 5 hold sample by sample; here the reference term $\Phi_X(H_P)$ is genuinely nonzero and the reference-sufficiency condition is active. Distributional statements are open.
- *Liouvillian and many-body:* the vectorized Liouvillian (or the Fock-space Hamiltonian) is a matrix on a Hilbert space with a chosen cell register, so Φ_Q, Φ_X are defined verbatim [7, 5]. What is *not* used or proved is any compatibility of the free operations with the complete-positivity structure of the Liouvillian.
- *Exceptional points:* $b_\lambda \rightarrow \infty$ as an eigenpair coalesces, so set $\Phi_Q = +\infty$ at defective points (lower semicontinuous extension). For degenerate but diagonalizable spectra, $\|P\| = 1/\cos \theta_{\text{max}}$ with θ_{max} the largest principal angle between the right and left eigenspaces [15]; a cell-density version of the degenerate case is open (Open problem 3).

10 Numerical verification

The script `nhse_qrt_v11_checks.py` tests every quantitative statement. Summary of results (double precision; HN at large L uses the closed form of Theorem 6(i), which involves no ill-conditioned eigenproblem):

- C1: direct eigensolver $\Phi_Q = \Phi_X$ for HN to all printed digits; matches the asymptotic formula of Theorem 6(ii) to 10^{-9} at $L = 40$ ($g = 0.25$).
- C2: $|R_j/(e^{g(L-1)} L_j) - 1| \leq 4 \times 10^{-15}$; the classical Chernoff minimizer sits at $s = 1/2$ for all modes.

- C3: for $g = 0.25$, $\Phi_Q - [g(L-1) - \log(L+1)] = 0.7873, 0.7390, 0.7217, 0.7161$ at $L = 100, 400, 1600, 6400$, converging to $\log(\frac{1}{2} \coth g) = 0.7137$ (the $-\log L$ term is verified to $L = 6400$); same for $g = 0.7$ (limit -0.1896).
- C4: $|\langle \ell | r \rangle| \leq F_{\text{pin}} \leq \text{BC}$ in all trials on random banded $q = 2$ matrices.
- C5: a complex-symmetric banded Toeplitz matrix has $\Phi_Q = 0.0595 > 0$ and $\Phi_X = 10^{-16}$.
- C6: a random $q = 2$ block-circulant ring has $\Phi_Q = 0.093 > 0$ and $\Phi_X = 10^{-16}$.
- C7: for HN ($L = 16$) a random unitary keeps $\Phi_Q = 1.868$ but lowers Φ_X from 1.868 to 0.252; a cell permutation leaves Φ_X unchanged.
- C8: $\|S_\delta H_{g_1} S_\delta^{-1} - H_{g_2}\| = 6 \times 10^{-16}$; $\beta' = \beta + \frac{1}{2} \log(Z^p Z^q)$ to 4×10^{-13} ; $|\Delta\beta| \leq \log \text{cond } S = 3.80$; branch success $\log p \in [-7.09, -5.98]$ versus $-2\delta(L-1) = -7.60$.
- C9: counterexample of Prop. 4: R_{skin} goes $0 \rightarrow 0.144$.
- C10: first-order sensitivity saturates εe^b .

Still required for a full numerical program (not yet done): tests of Theorem 7 on non-Hermitian SSH-type multiband rules, disordered ensembles, and the Blackwell criterion on HN modes.

11 Claims ledger and open problems

Claim	Status
Budgets, chain, faithfulness, reciprocity, translation-invariant ring free (Thm. 1)	Proved
Monotonicity, closure, sufficiency necessary (Thm. 2, Prop. 4, Thm. 4)	Proved
Covariant unitaries = cell permutations $\times$ on-site unitaries	Proved
Atlas-level complete conversion criterion (Thm. 3)	Imported (Blackwell)
Real-space extensivity (Thm. 5)	Proved
HN exact asymptotics, Chernoff optimality (Thm. 6, Prop. 2)	Proved, numerically confirmed
Dressing cost on HN ray (Thm. 8)	Proved
GBZ representation (Thm. 7)	Conditional on Assumption 2
Toeplitz faithfulness $r_{\text{skin}} > 0 \iff \text{winding} \neq 0$ (Conj. 1)	Conjecture
Critical exponents, disorder distributions, Liouvillian CP structure	Open

Open problem 1 (Dynamical-level monotone). *Define free supermaps on the instruments $\mathcal{I}_H^t$ (not on the atlas) and find a monotone that reduces to Φ_X on the atlas. The present theory does not provide this.*

Open problem 2 (Downhill HN conversion by free maps). *Decide, via Theorem 3, for which $g_1 > g_2 > 0$ the HN atlas of g_1 converts to that of g_2 under a common covariant stochastic map without dressing.*

Open problem 3 (Degenerate spectra). *Define Φ_X for degenerate eigenvalues using spectral projectors and prove the analogues of Theorem 1.*

Open problem 4 (Converse and critical sectors). *Prove Conjecture 1; extend to $q > 1$; classify χ_{skin} for scale-free models.*

12 Conclusion

The construction is a quantum resource theory of the NHSE in the following defined sense: an object class (boundary-paired biorthogonal atlases on a register of cells, any geometry), a free-operation class defined without reference to the resource value (boundary-blind, position-covariant, reference-sufficient), a quantum monotone Φ_Q and a geometric monotone Φ_X built from one fidelity divergence, free objects characterized by reciprocity and translation invariance, a complete atlas-level conversion criterion, an extensivity theorem valid without translation invariance, an exactly solved calibration, and an exact conversion cost on the HN ray. The remaining gaps are stated as a conjecture and open problems rather than absorbed into the claims.

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